

Improving Blood Pressure Control in a Pharmacist-Managed Hypertension Clinic

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Objective. To determine whether pharmaceutical care provided by a pharmacist-managed hypertension clinic results in better treatment outcomes when compared with traditional health care from a primary care physician.

Design. Prospective, controlled study.

Setting. Veterans Affairs Medical Center, Philadelphia, Pennsylvania.

Patients. Fifty six patients with essential hypertension; 27 were randomly assigned to the intervention group and 29 to the control group.

Intervention. Patients in the intervention group were scheduled monthly to meet with a clinical pharmacist who made appropriate changes in prescribed drugs, adjusted dosages, and provided drug counseling in accordance with the hypertension guidelines in the sixth report of the Joint National Committee on the Detection, Evaluation, and Treatment of High Blood Pressure (JNC VI). Patients in the control group received standard care from their physicians. The study period was 6 months.

Measurements and Main Results. Treatment outcomes were measured by changes in compliance, blood pressure, and patient satisfaction. The Short Form-36 health survey and a patient satisfaction survey were used to measure changes in patient satisfaction, and a compliance evaluation survey measured compliance. Twenty-one (81%) patients in the intervention group attained their blood pressure goal of below 140/90 mm Hg at the completion of the study versus only eight (30%) in the control group ($p < 0.0001$). Of 11 patients with diabetes in the intervention group, 10 (91%) attained their blood pressure goal ($< 130/80$ mm Hg) versus only two (12 %) of 16 patients with diabetes in the control group ($p < 0.0001$). No significant differences in patient satisfaction or compliance were reported between the intervention and control groups.

Conclusions. Pharmaceutical care improves blood pressure control and results in more patients with hypertension reaching their blood pressure goal.

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Hypertension is a chronic disease that affects approximately 50 million American adults, one third of whom are unaware of their condition. The disease has been appropriately nicknamed the silent killer because most patients with hypertension have no symptoms, and therefore the disease often goes undetected. Even among patients diagnosed with hypertension, noncompliance and inappropriate drug therapy often

interfere with effective treatment. If not properly controlled, hypertension can lead to blindness, renal failure, heart disease, and stroke.^{1,2}

Despite the decline in heart disease over the last 30 years, it still remains the leading cause of death in the United States.^{1,2} Since hypertension remains one of the modifiable risk factors for heart disease, effective treatment is vital. The third National Health and Nutrition Examination Survey was conducted from 1988–1991 through in-home interviews and clinical examinations of 9901 people across the country. The survey reported that hypertension is adequately controlled in only 24% of all patients diagnosed with the disease. Noncompliance with prescribed drug regimens is believed to account for much of this discrepancy.¹

Noncompliance among all patient groups has long been a primary concern of medical and pharmaceutical professionals. The most significant factors in noncompliance are a lack of knowledge about prescribed drugs and their side effects, and a lack of understanding about the consequences if a prescribed drug regimen is not followed. Offering disease-specific management education programs to patients has been highly effective in improving compliance with drug therapy.³

Previous studies conducted at Veterans Affairs Medical Centers across the country have demonstrated that clinical pharmacy services can have a significant impact on compliance and patient satisfaction, and thus on drug therapy outcomes.^{4–8} We conducted this study to determine whether a pharmacist-managed hypertension clinic improves treatment outcomes in patients with hypertension.

Methods

In this prospective, controlled, 6-month study, treatment outcomes were measured by changes in compliance, blood pressure measurements, and patient satisfaction between the start and end of the study. The study was conducted at the Veterans Affairs Medical Center in Philadelphia, Pennsylvania; the site was chosen because over 55% of its patients have hypertension, and the center's health care providers identified noncompliance as a primary reason for poor treatment outcomes.

Patient Selection

After the investigational review board approved the study, all patients who came to the general

medicine clinic for management of their hypertension during the 1-month enrollment period were identified as potential candidates for the study. A total of 156 patients, both men and women, were screened for eligibility; 56 patients with a confirmed diagnosis of hypertension were considered eligible and were invited to participate. All patients enrolled in the study had stage 1, 2, or 3 hypertension as defined by the eligibility criteria and by the sixth report of the Joint National Committee on the Detection, Evaluation, and Treatment of High Blood Pressure (JNC VI).² Inclusion criteria were age older than 18 years, confirmed diagnosis of essential hypertension (defined as systolic blood pressure > 140 mm Hg or diastolic blood pressure > 90 mm Hg), receiving antihypertensive drug therapy (and blood pressure > 140/90 mm Hg), receiving all drugs from a Veterans Affairs Medical Center pharmacy (for assessment of compliance and drug interactions), and not receiving care at the pharmacist-managed clinic (until the study began).

Patients were excluded if they had a secondary cause of hypertension, such as chronic renal disease, renovascular disease, pheochromocytoma, Cushing's syndrome, and primary aldosteronism; had missed more than three appointments in the last year; or were in hypertensive crisis (defined as systolic blood pressure > 210 mm Hg or diastolic blood pressure > 110 mm Hg). Patients also were excluded if they had a diagnosis of New York Heart Association class III or IV chronic heart failure, end-stage renal disease, a psychiatric disorder, severe hepatic dysfunction defined as transaminase levels greater than 3 times the upper limit of normal, or terminal cancer or other condition that limited life expectancy to less than 1 year.

The nature of the study was described to the patients, and informed consent forms were signed by those eligible and interested in participating.

Patients were randomly assigned to either the intervention group (27 patients) or the control group (29 patients). Patients in the intervention group were scheduled to see the clinical pharmacist once/month at the pharmacist-managed hypertension clinic. This pharmacist, who had prescribing authority at the medical center, made appropriate drug therapy changes (in both drug selection and dosage) for blood pressure control in accordance with JNC VI.² The pharmacist did not make any changes in their patients' other drugs that may adversely

affect blood pressure, such as venlafaxine and sibutramine. Drug counseling, consisting of a thorough discussion about side effects, recommended lifestyle changes, and an assessment of compliance, was provided at each visit. Patients were allowed to receive care for comorbid conditions from their primary care providers during the study period, but these providers could not make changes in the patients' antihypertensive drug regimens.

Patients in the control group received traditional pharmacy services (dispensing, brief counseling about drugs, and review of drug profiles) but did not make monthly visits to the pharmacist-managed hypertension clinic. They received care from their primary care providers as needed, at least once/year. All primary care providers were informed that their patients were enrolled in the study.

Intervention

At the start and end of the 6-month period, all 56 patients had their blood pressures measured in both arms while seated, after at least 5 minutes of rest. Measurements were obtained by a clinical pharmacist using an auscultatory sphygmomanometer. The general goal was a blood pressure measurement below 140/90 mm Hg. For patients with diabetes, the goal was below 130/80 mm Hg.

In addition, patients filled out a demographic information form and three questionnaires. The first was a compliance evaluation that assessed the degree to which patients followed their prescribed drug regimen as reflected by self-reporting. This evaluation was not previously administered or validated. The second questionnaire was a patient satisfaction survey³ that reflected patients' assessments of their interaction with the clinical pharmacist and their overall satisfaction with the service received at the pharmacist-managed clinic. The third questionnaire was the Medical Outcomes Study Short Form-36 health survey,⁹ which measured quality-of-life indicators such as restrictions on employment and participation in stress-inducing activities.

Compliance

For the intervention group, compliance was measured at each monthly visit; the control group was assessed at baseline and at the end of the study. In addition to the compliance evaluation filled out by the patient, records of

drug refills were obtained from the pharmacy. Noncompliance was defined as missing more than three doses of drug in 1 week, or having pharmacy records indicate failure to refill drugs within 2 weeks after the scheduled refill date.

Statistical Analysis

Outcome measurements were derived from statistical analysis of data obtained from the 26 intervention and 27 control patients. The paired *t* test was used to measure differences in blood pressure between baseline and end of study. A two-sample *t* test was used to assess the significance of the treatment difference between the mean systolic and diastolic blood pressure change from baseline to end of study.

This method of analysis assumes that the data are distributed symmetrically. We checked for significant violations of this assumption and validated the results with a Wilcoxon sum rank test, which requires no parametric assumption. Fisher's exact test was used to compare the number of patients reaching their blood pressure goal in each group, as well as the number of patients with diabetes who reached their goal. Fisher's exact test also was used to compare the frequencies between drug therapies used for treatment of hypertension.

Categorical data, such as compliance and quality of life, were analyzed with the Pearson χ^2 test. Because of the low cell sizes in the contingency tables for the patient satisfaction data, the Fisher's exact test was used. This procedure gives exact results for 2 x 2 tables in which the standard χ^2 is not applicable.¹⁰

With the quality-of-life scores on the Short Form-36 health survey, comparisons between the intervention and control groups were evaluated by two-sample *t* tests at baseline and after intervention, and the difference between baseline and end of study was calculated. These results were validated with a nonparametric Wilcoxon rank sum test.¹⁰ All analyses used SAS (version 8, SAS Institute, Cary, NC).¹¹ The significance level was 0.05 for all statistical tests.

Results

Demographics

A total of 53 of 56 patients completed the study. One patient in the intervention group withdrew because he could not comply with monthly visits to the pharmacist clinic. One patient in the control group withdrew from the

Table 1. Baseline Demographics and Clinical Characteristics

	Intervention Group	Control Group
M/F	26/0	27/0
Age, mean $\pm$ SD (yrs)	64 $\pm$ 10.9	65.5 $\pm$ 7.8
Weight, mean $\pm$ SD (kg)	204.57 $\pm$ 44.6	204.63 $\pm$ 43.7
Patients with diabetes, no. (%)	11 (42)	16 (59)
Smokers, no. (%)	4 (15)	7 (26)
Race, no.		
African-American	22	19
Caucasian	3	7
Other	1	1

Table 2. Changes in Blood Pressures Between Baseline and End of Study

Blood Pressure	Intervention Group (n=26)	Control Group (n=27)	p Value
Systolic (mm Hg)			
Baseline	149.0 $\pm$ 15.3	152.8 $\pm$ 14.3	0.252
End of study	130.5 $\pm$ 13.2	148.4 $\pm$ 21.0	0.0002
Diastolic (mm Hg)			
Baseline	89.8 $\pm$ 10.9	77.9 $\pm$ 11.9	0.0012
End of study	77.5 $\pm$ 10.7	80.4 $\pm$ 11.4	0.259

Data are mean $\pm$ SD.

Table 3. Changes in Blood Pressures of Patients with Diabetes

Blood Pressure	Intervention Group (n=11)	Control Group (n=16)	p Value
Systolic (mm Hg)			
Baseline	152.18 $\pm$ 11.47	148.81 $\pm$ 7.21	0.278
End of study	132.2 $\pm$ 11.00	147.43 $\pm$ 8.65	0.0002
Diastolic (mm Hg)			
Baseline	87.27 $\pm$ 9.88	74.06 $\pm$ 10.55	0.0012
End of study	74.2 $\pm$ 10.2	78.43 $\pm$ 5.22	0.254

Data are mean $\pm$ SD.

study because he relocated to a different state. Another patient in the control group withdrew because he no longer was interested in participating in the study. Of the 53 patients who completed the study, 41 (77%) were African-American, and 11 (21%) were cigarette smokers (Table 1). Although women were not excluded from the study, all patients enrolled were men, predominantly African-American.

Blood Pressure Control

Patients in the intervention group had higher mean diastolic blood pressures at baseline than those of patients in the control group ($p=0.0012$; Table 2). No other differences in baseline

characteristics between groups were detected.

Of the 26 patients in the intervention group, 21 (81%) attained a blood pressure below 140/90 mm Hg at the end of the study versus only eight (30%) of the 27 patients in the control group ($p=0.001$).

Mean changes in systolic blood pressure from baseline for the intervention and control groups were -18.4 (95% confidence interval [CI] -26.3, -10.5) and -3.98 (95% CI -11.8, 3.79), respectively ($p=0.01$). Mean changes in diastolic blood pressure from baseline for the intervention and control groups were -12.38 (95% CI -16.49, -8.28) and 2.54 (95% CI -1.49, 6.57), respectively ($p=0.001$). Of the 11 patients with diabetes in the intervention group, 10 (91%)

Table 4. Antihypertensive Drugs Prescribed During the Study

Drug Category	No. (%) of Patients		p Value
	Intervention Group (n=26)	Control Group (n=27)	
β -Blocker	10 (38)	12 (44)	0.79
Calcium channel blocker	16 (62)	14 (52)	0.44
Thiazide diuretic	16 (62)	17 (63)	1.00
Angiotensin-converting enzyme inhibitor	18 (69)	26 (96)	0.052
Angiotensin II receptor antagonist	0	1 (4)	1.00
α -Antagonist	6 (23)	9 (33)	0.55
Vasodilator	0	1 (4)	1.00

Table 5. Compliance Survey

Survey Question	% of Patients		p Value
	Intervention Group	Control Group	
1. How often do you forget to take your medication (forgets $\geq$ once/wk)?	68	48	0.252
2. How often do you stop taking your medication when you are feeling better? ($\geq$ once/wk)	32	20	0.520
3. How often do you stop taking your medication when you think it is making you feel worse ($\geq$ once/wk)?	40	20	0.217
4. When your medication does not seem to be working, how often do you take more than your health care provider prescribed ($\geq$ once/wk)?	8	8	1.00
5. Were you away from home overnight in the last 3 months (answered yes)?	52	40	0.571
6. If you answered yes to question 5, did you forget to take your medication when you were away from home overnight? ^a	15	10	1.00
a. Never (0 times/wk)			
b. Rarely (< twice/wk)			
c. Sometimes (2–3 times/wk)			
d. Always (> 3 times/wk)			

^aAnswers to question 6 were the percentages of patients who selected answers c and d.

attained the goal blood pressure of below 130/80 mm Hg versus only two (12 %) of 16 patients with diabetes in the control group ($p=0.001$; Table 3).

More patients in the control group than the intervention group received angiotensin-converting enzyme inhibitors ($p=0.052$; Table 4). No significant differences were found in the prescribing of thiazide diuretics, β -blockers, calcium channel blockers, angiotensin II receptor antagonists, direct vasodilators, or α -blockers ($p=0.44$).

Compliance

No significant difference in compliance was reported between ($p>0.25$) or within ($p>0.07$) the two groups at baseline or end of study (Table 5). In the intervention group, 68% of patients admitted forgetting to take their drug at least once/week versus 48% in the control group ($p=0.252$). Over 90% of patients in both groups

stated that they took their drugs as directed by their health care professional and did not take more than prescribed ($p=1.00$).

Pharmacy records indicated that 22 (85%) of the 26 patients from the intervention group received their refills for antihypertensive agents within 2 weeks of the next scheduled refill date, versus 25 (93%) of the 27 patients from the control group ($p>0.42$). Patients in the intervention group visited the clinical pharmacist once/month as scheduled, but only 20 (77%) had an appointment with their physician at least once during the study period. Only 17 (63%) patients in the control group had an appointment with their primary care provider during the 6-month study period ($p=0.372$).

Patient Satisfaction and Quality of Life

Fewer patients in the intervention group complained of distractions in the pharmacy or clinic compared with the control group ($p=0.018$;

Table 6. Patient Satisfaction Survey

Survey Item	% of Patients ^a		p Value
	Intervention Group	Control Group	
The pharmacist spends as much time with me as necessary.	72	46	0.063
The pharmacist explains the possible side effects that a new medication may cause.	58	36	0.117
If I have a question about my prescription, a pharmacist is always available to help me.	64	48	0.255
The pharmacist knows how to explain things in a way that I understand.	79	63	0.204
The pharmacist is not as thorough as he/she should be.	14	30	0.175
The pharmacist makes sure that I understand how to take my take my medication.	75	74	0.932
Sometimes the pharmacist does not spend enough time with me.	4	17	0.129
The pharmacy/clinical staff are friendly.	88	70	0.116
The pharmacy/clinical staff look professional.	84	77	0.559
I have to wait a long time before I get to see the pharmacist.	20	32	0.354
I have trouble understanding the pharmacist.	12	23	0.329
The pharmacy/clinical staff seems to have a genuine interest in me as a person.	68	55	0.344
There are so many distractions in the pharmacy/clinic area that I can't get good service.	8	36	0.018
I am very satisfied with the pharmacy services that I receive.	88	68	0.098

^aPercentages of patients who experience the situation most of the time. Data for sometimes, very rarely, and never are not shown.

Table 7. Health-Related Quality-of-Life Scores

Domain	Baseline			At 6 Months		
	Intervention Group	Control Group	p Value	Intervention Group	Control Group	p Value
Health perception	39.35 ± 15.62	39.21 ± 13.55	0.9090	41.98 ± 14.35	41.8 ± 15.74	0.9670
Physical functioning	65.33 ± 25.87	63.33 ± 29.21	0.7988	65.97 ± 27.7	63.043 ± 28.90	0.7243
Role limitations, physical	53.846 ± 44.55	67.308 ± 44.59	0.2814	50 ± 45.64	56.52 ± 43.44	0.6152
Role limitations, emotional	56.56 ± 48.56	70.37 ± 44.42	0.2704	44 ± 46.34	56.52 ± 48.39	0.364
Social functioning	65.833 ± 35.13	68.148 ± 33.86	0.8118	63.33 ± 33.71	67.5 ± 37.21	0.6862
Energy, fatigue	60.88 ± 21.52	67.24 ± 21.47	0.2947	59.87 ± 18.01	68.2 ± 18.53	0.1141
Bodily pain	59.40 ± 30.94	75.31 ± 23.54	0.0397	58.67 ± 27.87	70.67 ± 29.20	0.1437

Scores range from 0–100.

Table 6). Although no statistical differences in patient satisfaction were noted, more patients in the intervention group than in the control group felt that the pharmacist spent more time with them during their visit ($p=0.06$). In the intervention group, 88% were satisfied with pharmacy services versus only 68% in the control group ($p=0.098$). No significant changes in patient satisfaction were noted in either group from baseline to end of study ($p>0.1$).

On the quality-of-life assessment, no statistically significant differences were noted between the two groups from baseline to end of study (Table 7). However, patients in the control group reported higher scores for bodily pain at baseline than patients in the intervention group ($p<0.04$). No significant difference was reported within the two groups at baseline or end of study ($p>0.2$).

Discussion

Patients in the intervention group, who received care in the pharmacist-managed clinic, showed significant improvement in blood pressure management compared with patients in the control group, who were managed by their physician. This treatment outcome may be due to the frequent follow-up care and patient education provided by the pharmacist.

Most patients in this study were African-Americans, among whom the prevalence of hypertension in the United States is the highest. The incidence of heart disease, stroke, and renal disease also is greater among African-Americans than the general population.^{1, 2, 12, 13} Yet research on effective treatment of hypertension in African-Americans is lacking, despite studies suggesting that African-Americans often require a different

approach to drug therapy than do Caucasians.^{2, 14,}

¹⁵ Studies have suggested that African-Americans with hypertension are more responsive to diuretics and calcium channel blockers than other antihypertensive agents.^{13, 16, 17}

This study had inadequate power to detect a significant difference in compliance, health perception, and patient satisfaction. Although no difference in these factors was found, compliance may have been greater in the intervention group. Monthly assessment of compliance may have resulted in less recall bias because these patients were more likely to remember how often they forgot to take their drugs than patients in the control group, whose compliance was assessed only at baseline and end of study. Pharmacy records showed no significant difference in compliance between the groups.

Patients in the control group reported more bodily pain at baseline; however, they had lower mean blood pressures than patients in the intervention group at baseline. Both groups were pleased with the overall service provided by the pharmacy department.

This study was conducted in only one clinical setting, which limits the ability to extrapolate the results to other settings. The patient satisfaction forms were filled out in the pharmacy clinic, which may have resulted in higher scores. Only three clinical pharmacists measured blood pressure. However, because it was not possible to conduct a blinded study, the physicians may have altered their usual practice habits.

Although women were not excluded from the study, the patients enrolled were all men and predominantly African-American. The study did not assess complications resulting from hypertension, such as coronary heart disease, stroke, and retinopathy.

Physicians may not have enough time to provide the level of education, drug counseling, and follow-up care required by many patients to ensure that their drug regimen is providing adequate blood pressure control without adverse effects. Scheduling difficulties result in a lack of continuity of care, and patient dissatisfaction leads to noncompliance; the result is ineffective treatment.

The improvement of blood pressure management in the intervention group is likely due to several variables but could be due to the amount of time spent with patients at each visit. Although time was not measured specifically, most pharmacist visits lasted 30–45 minutes versus only 20 minutes allotted for physician visits.

Conclusion

Given the prevalence and health risks posed by hypertension, establishing a means to reduce the frequency of this disease has far-reaching implications. This study demonstrates that pharmaceutical care provided by a pharmacist-managed hypertension clinic improves blood pressure control and results in more patients with hypertension reaching their goal blood pressure measurement.

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